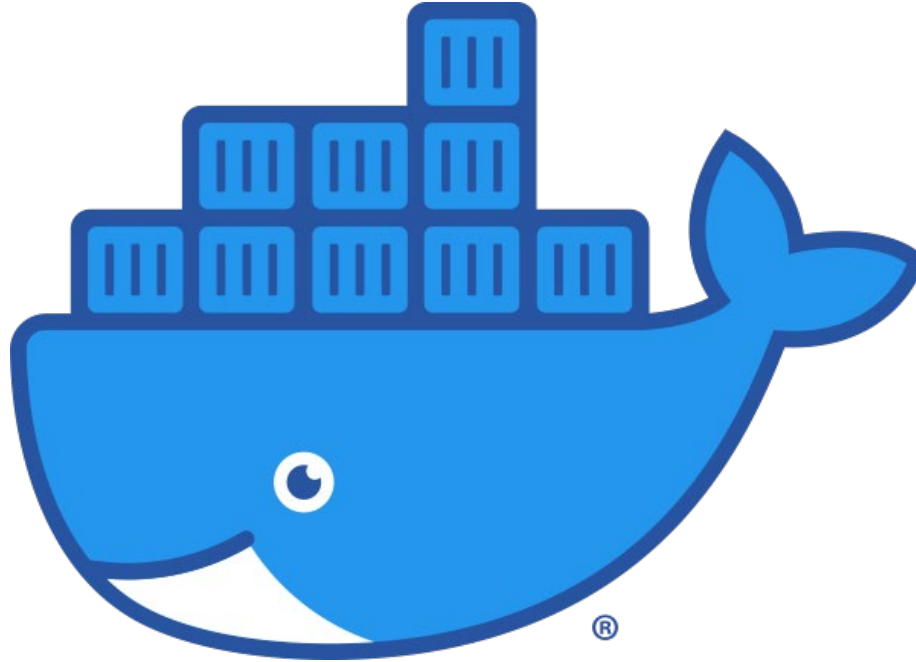




Docker Swarm Step-by-Step

Agenda

Docker Overview

Docker Objects Overview

Why orchestration?

Docker Swarm

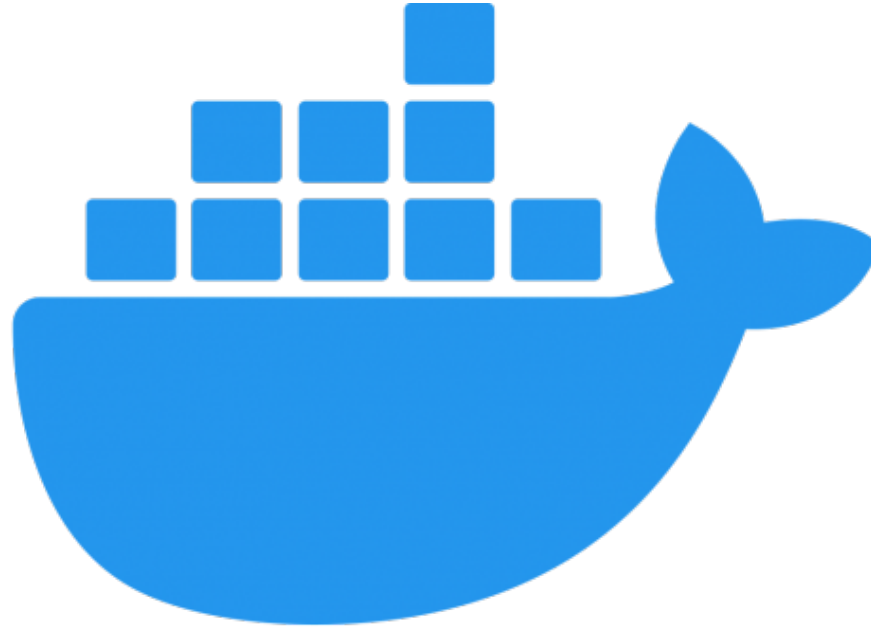
Docker Swarm Architecture & Components

Docker Swarm Common commands

Deploying in a Docker swarm

Managing a Docker swarm

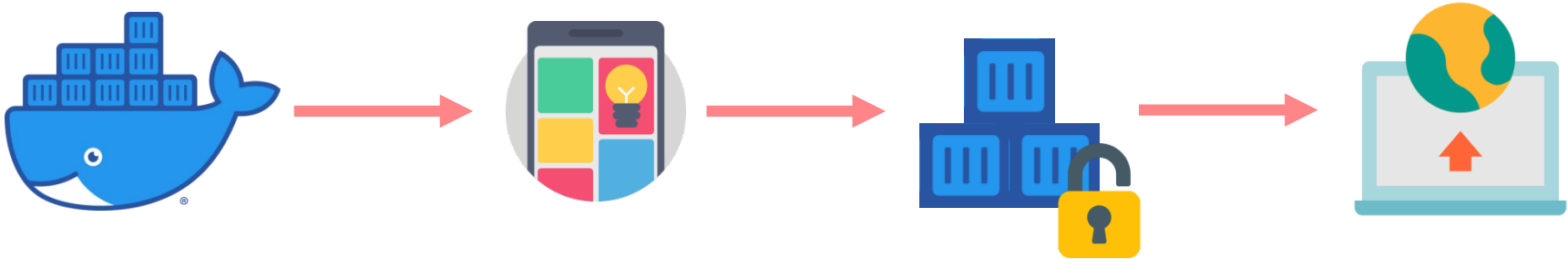
Q/A



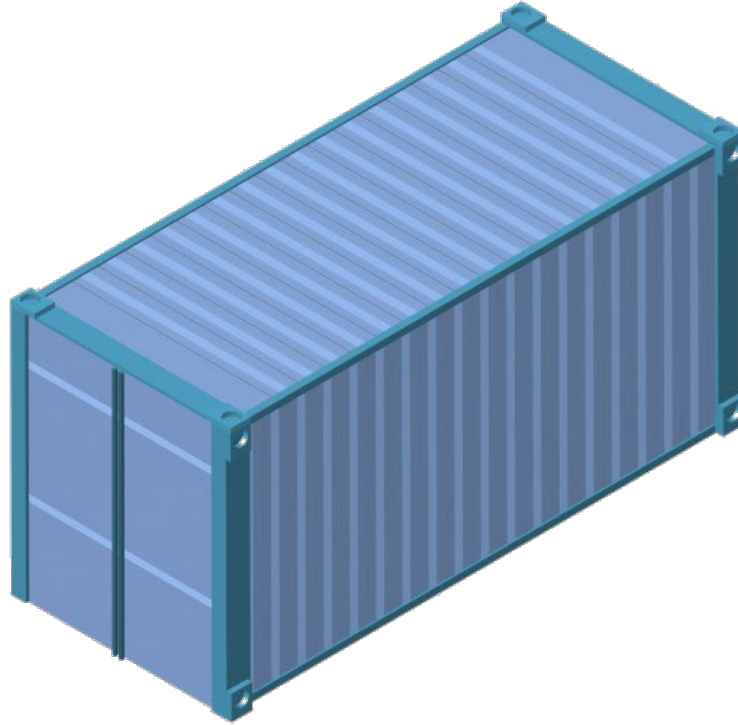
What is Docker?

What is Docker?

Docker is a tool that helps in developing, building, deploying and executing software in isolation. It does so by creating containers that completely wrap a software.



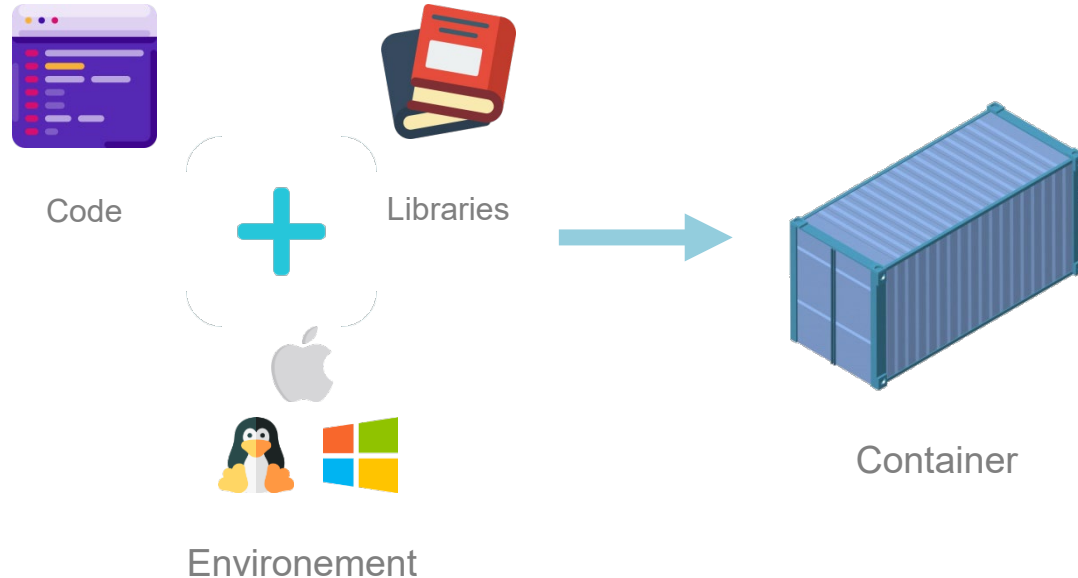
The Isolation provided by container gives a layer of security to the containers.



What is a **Container**?

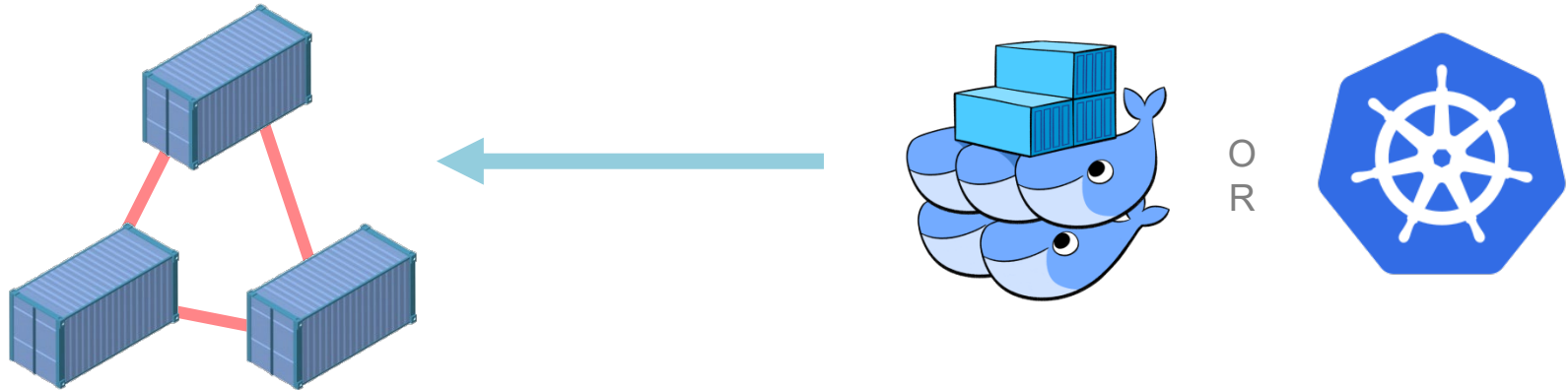
What is a Container?

Containers are software that wrap up all the parts of a code and all its dependencies into a single deployable unit that can be used on different systems and servers.



What is a Container?

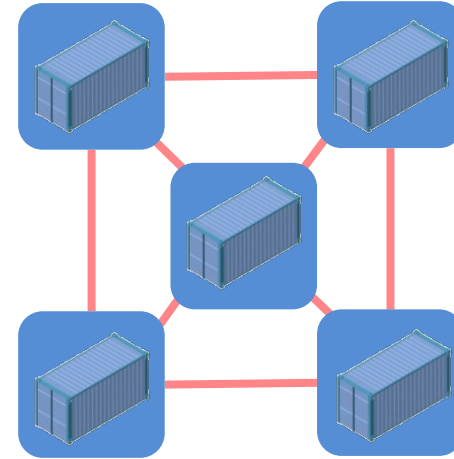
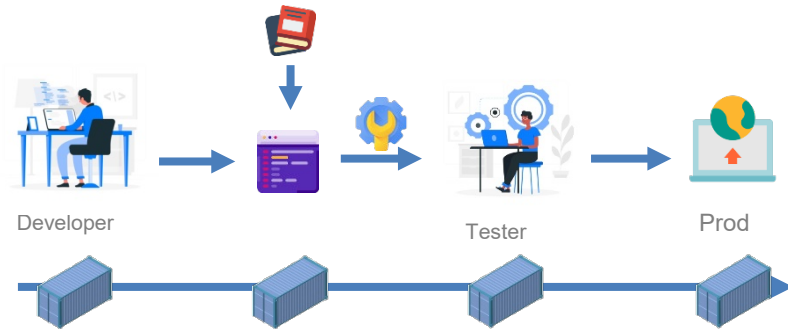
Multiple isolated containers can be launched together to form Microservices which can be easily managed using any orchestration tool e.g. Docker Swarm, Kubernetes, etc.



Why do we need Containers?

Consistent Development
Environments

Microservices



Docker Environment



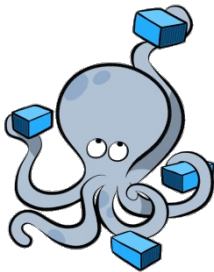
Docker Engine



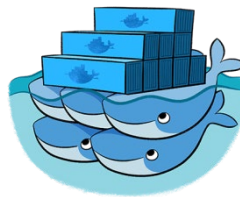
Docker Objects



Docker Registry



Docker Compose



Docker Swarm

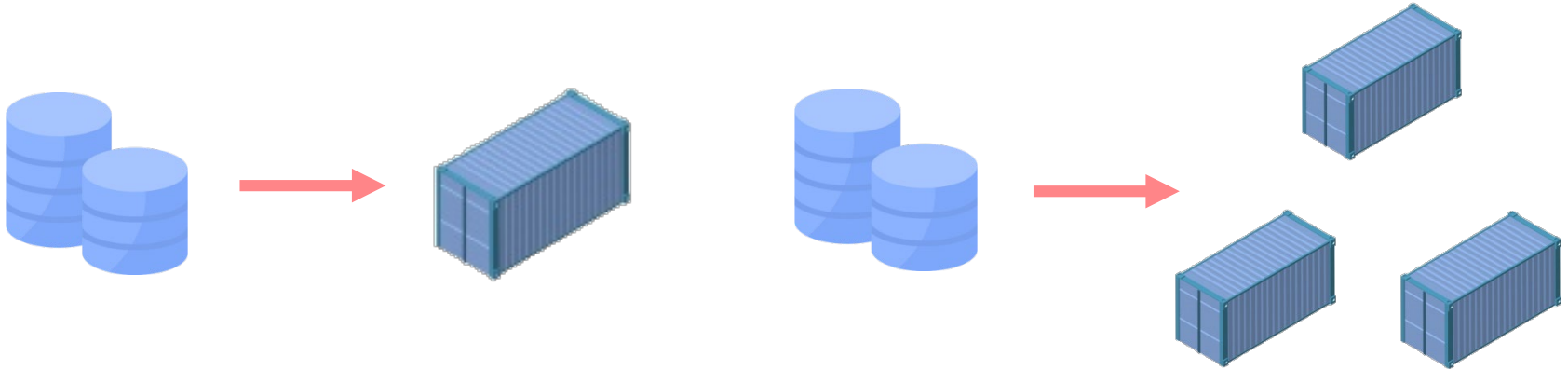
Docker Objects - Images

Docker images are sets of instructions that are used to create containers and execute code inside it.



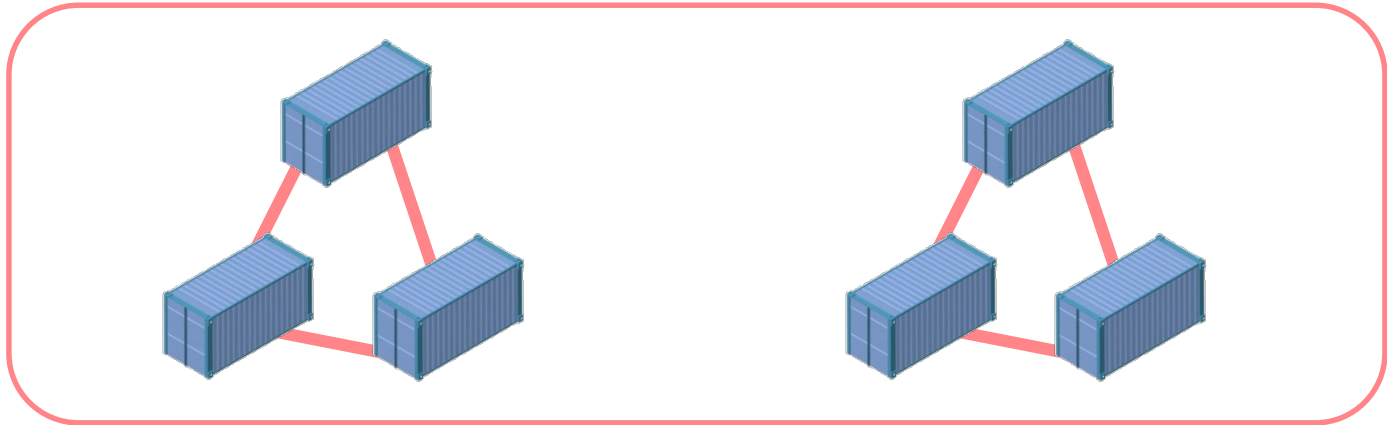
Docker Objects - Volumes

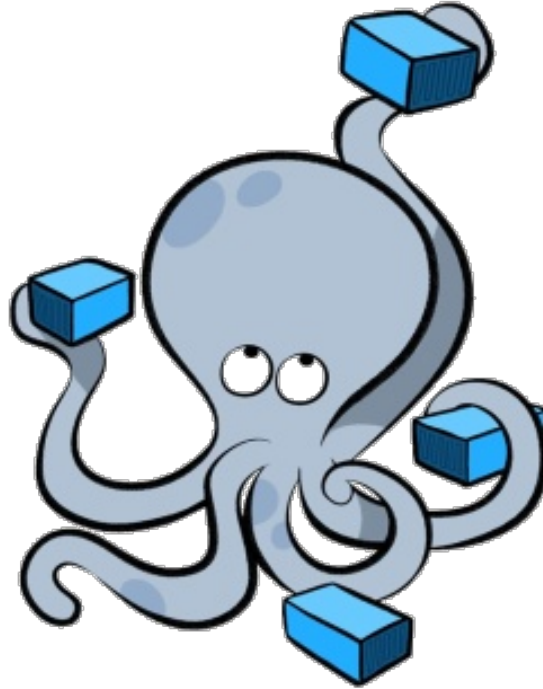
Docker Volumes are basically persistent storage locations for the containers. They can be easily & safely attached and removed from different container. And they are also portable from system to another.



Docker Objects - Network

A Docker network is basically a connection between one or more containers. One of the more powerful things about the Docker containers is that they can be easily connected to one other and even other software, this makes it very easy to isolate and manage the containers

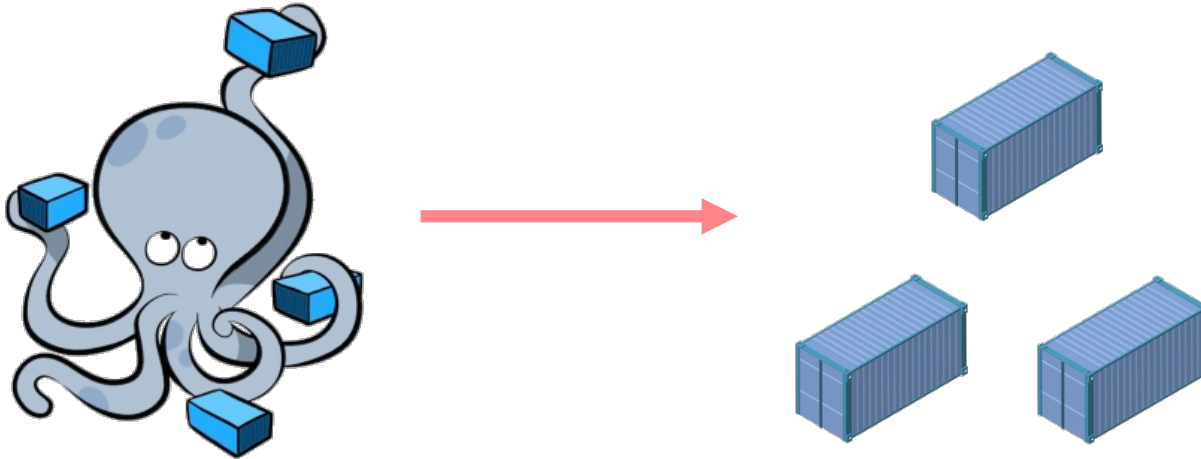




Docker Compose

Docker Compose

For Now let's just understand that Docker Compose is just a Service within Docker that let's us launch multiple containers at the same time.

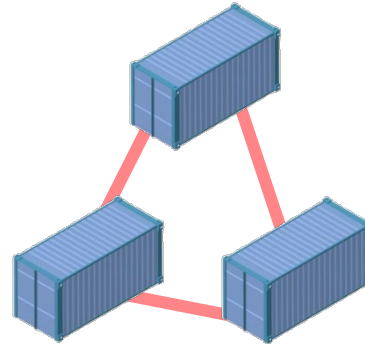




Why Orchestration?

Why Orchestration?

Orchestration allows us to manage and maintain multiple containers. This is especially helpful in software development where we maybe making use of micro-services architecture as it breaks down the software into small manageable chunks. Having different configurations and environments becomes easier with orchestration.



Benefits of Orchestration?

Easy Deployment

Creates a layer of security

Easy management

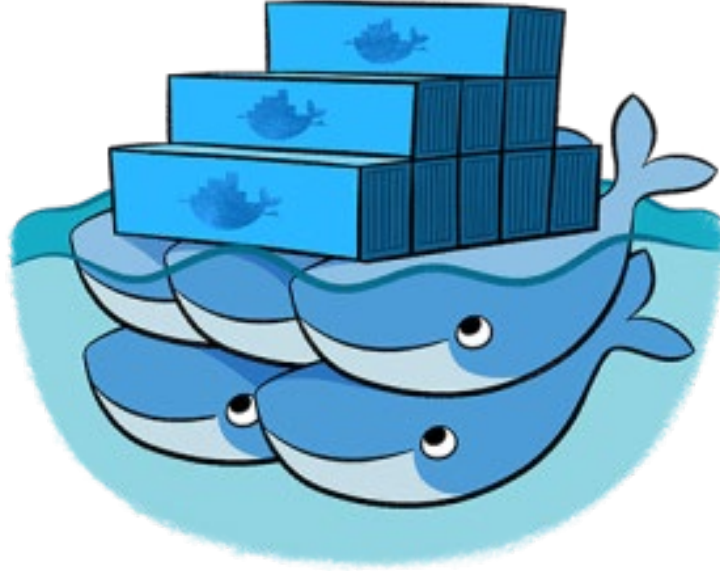
Easy resource management

Allows for health monitoring of the containers

Load balancing among different containers

Easy updating

Easy scaling up and rolling back



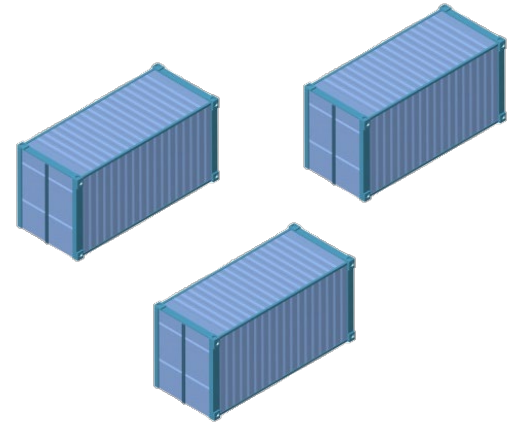
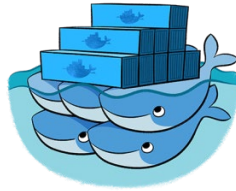
Docker Swarm

Docker Swarm

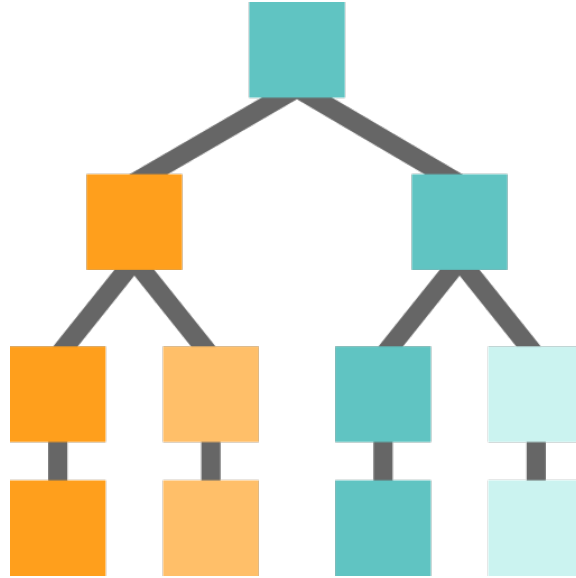
Docker Swarm is an orchestration service within Docker that allows us to manage and handle multiple containers at the same time. It is also a cluster of multiple containers.



Manager Node

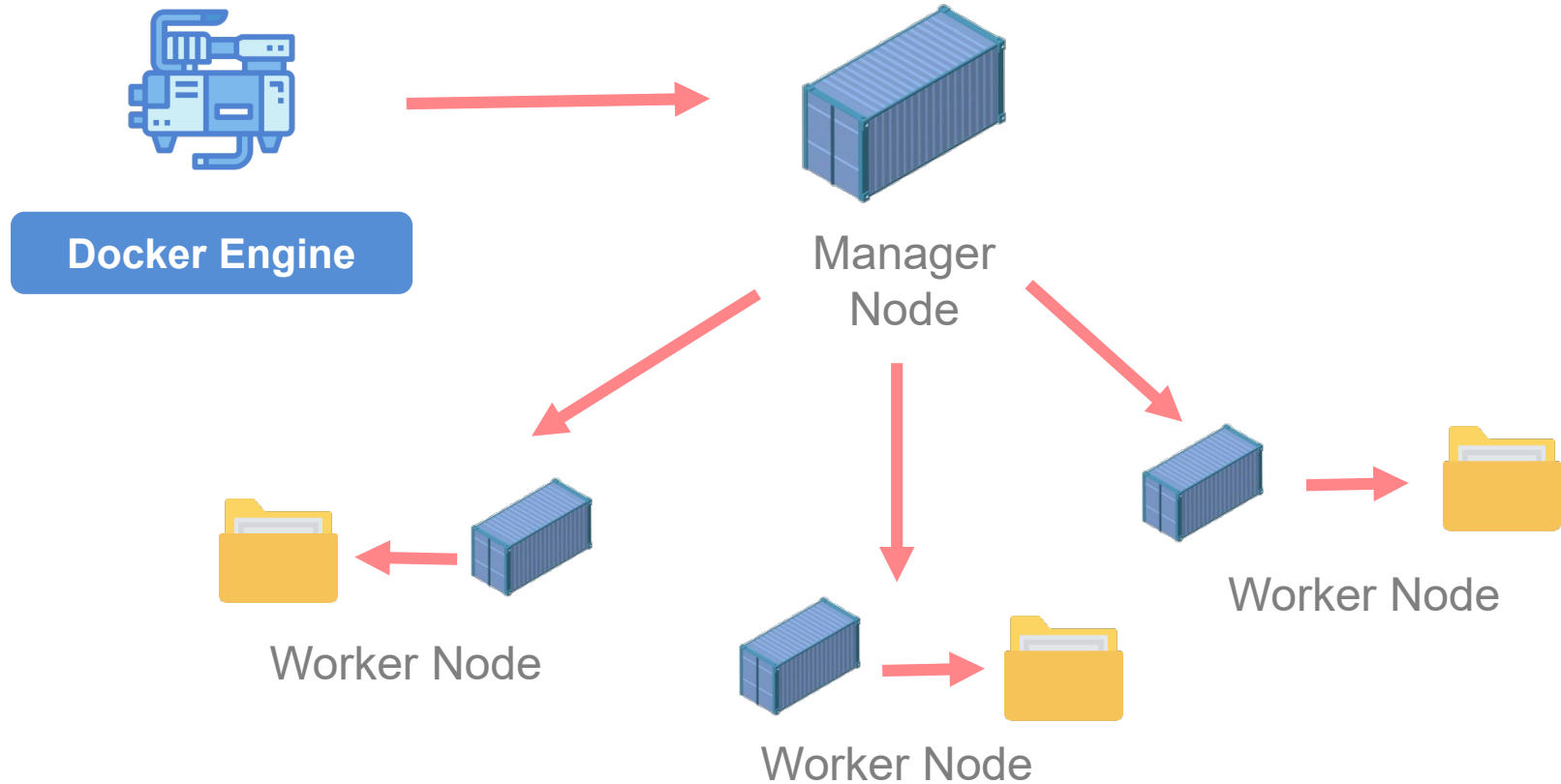


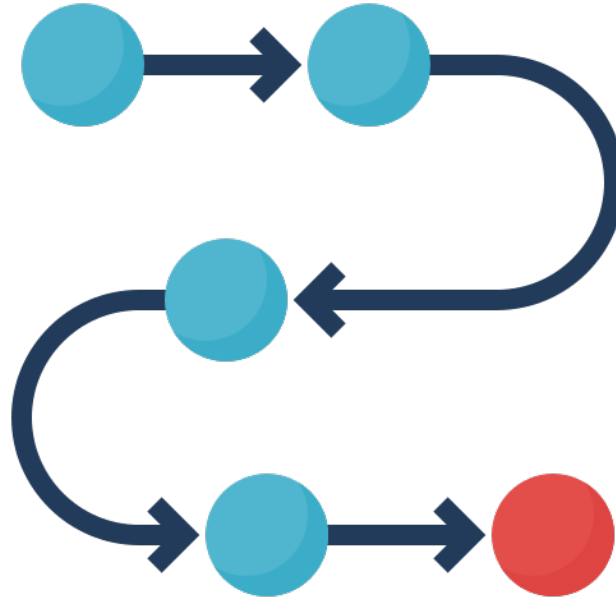
Worker Node



Docker Swarm Architecture

Docker Swarm architecture





Docker Swarm Components

Docker Swarm Components



Service



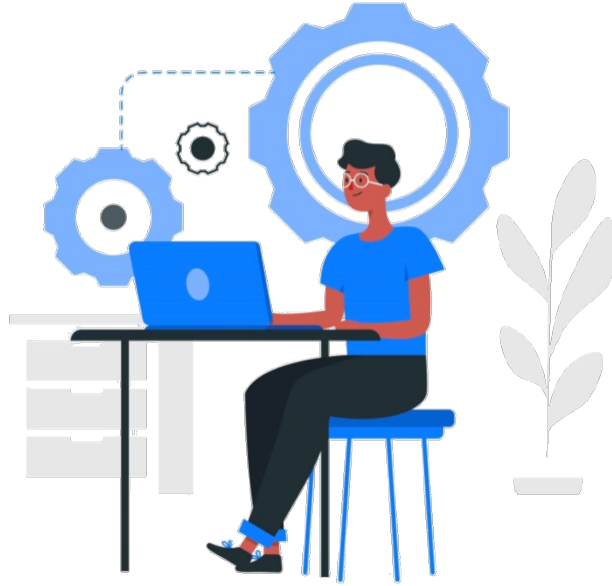
Task



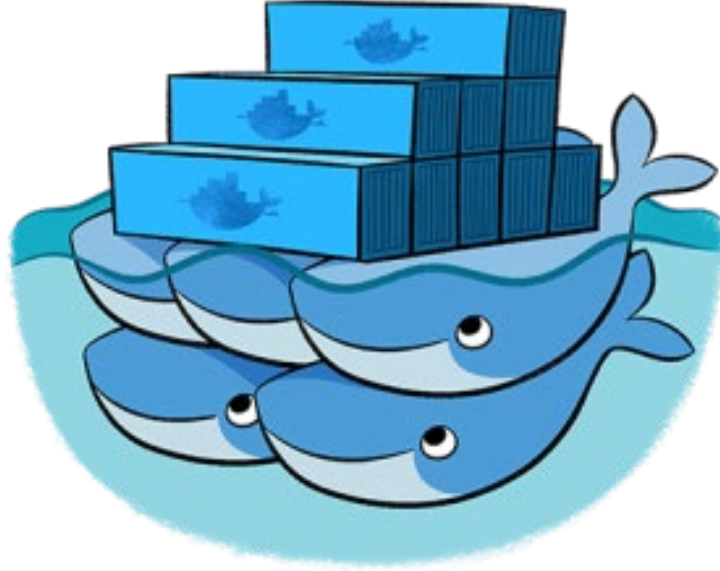
Manager



Worker



Docker Swarm **Common Commands**



Deploying in a Docker Swarm



Managing a Docker Swarm



Q&A

